

Technique Packet No. 2

Working Backwards — how to read a mind, and other tricks.

Name: _____ Date: _____

Here is a trick you can play on a friend. Tell them: think of a secret number — don't tell me. Add 6 to it. Now double what you have. What did you get? They say "20," and after a moment you announce their secret number: **4**. They will be amazed. How did you know?

You didn't guess. You walked their steps **backwards**. They ended at 20. The last thing they did was double, so undo it — halve 20 to get 10. Before that they added 6, so undo that too — subtract 6 to get 4. That is their number. The secret behind every "I can read your mind" number trick is the same: **when you know where someone ended up but not where they began, run the steps in reverse and undo each one.**

Forward — do the steps



Backwards — undo them, last step first

Same chain, two directions: doing the steps runs left to right; undoing them runs right to left, last step first.

THE TECHNIQUE — Working Backwards.

When a problem tells you the END but not the START:

1. Write down the steps that were done, in order.
2. Undo the **last** step first, using the opposite operation (undo + with −, × with ÷, and so on).
3. Keep undoing, one step at a time, until you reach the start.

When to reach for it: whenever you know how something finished but not how it began — “find the number I started with,” a process run in reverse, an amount before some changes were made. Don't guess and check — undo.

Watch how it works

WORKED EXAMPLE 1

I think of a number, add 6, then double it, and get 20. What was my number?

1. The steps, in order: start with the number, add 6, then double.
2. Undo the last step first. The last step was doubling, so halve: $20 \div 2 = 10$.
3. Undo the step before — it was adding 6 — so subtract: $10 - 6 = 4$.
4. So my number was **4**. (Check forwards: $4 + 6 = 10$, doubled is 20. ✓)

WORKED EXAMPLE 2 — when it doesn't look like a “start” problem

A candle burns down to half its height every hour. After 3 hours it is 4 cm tall. How tall was it to begin with?

1. This is working backwards in disguise: we know the end (4 cm, after 3 hours) and want the start.
2. Each hour the height was halved, so to undo one hour, double.
3. Undo hour 3: $4 \times 2 = 8$. Undo hour 2: $8 \times 2 = 16$. Undo hour 1: $16 \times 2 = 32$.
4. So the candle started at **32 cm**. (Check: $32 \rightarrow 16 \rightarrow 8 \rightarrow 4$. ✓)

WORKED EXAMPLE 3 — undo in reverse order

A girl gives away half her stickers, then gives away 3 more, and has 5 left. How many did she start with?

1. The steps, in order: start, give away half, then give away 3 more, leaving 5.
2. Undo the last step first. The last thing she did was give away 3, so add them back: $5 + 3 = 8$.
3. Before that she gave away half — so 8 is the half she kept. Undo by doubling: $8 \times 2 = 16$.
4. So she started with **16**. (Undoing in the wrong order — doubling first — would give the wrong answer.)

Now you

Show your undo steps — not just the final number.

1. I think of a number, subtract 4, then multiply by 3, and get 18. What was my number?
2. A jug of juice: you pour out half of it, then drink 1 more cup, and 2 cups are left. How many cups were in the jug at the start?
3. A sourdough starter doubles its weight every day. After 4 days it weighs 80 grams. How much did it weigh at the start?

4. I think of a number, add 5, then double it, then subtract 4, and get 20. What was my number?

5. A bus has some people on it. At the next stop 4 people get off and 7 people get on. Now there are 19 people. How many were on the bus before the stop?

6. Maya won a bet that tripled her money. Then she spent \$8. Now she has \$40. How much money did she start with?

Answer Key — for parents

What this packet teaches. One named technique — working backwards — shown first (a mind-reading hook, the chain picture, and three worked examples), then applied six times. The three worked examples are deliberately different *kinds* of the same move: a plain number-machine (1), a disguised problem that doesn't announce itself and undoes the same step three times (2), and one where the order of undoing matters (3) — so every kind of problem below has been modeled before they meet it. The lines are written one sentence each on purpose: they double as models of a written explanation.

Answers.

1. Undo last-first: $18 \div 3 = 6$, then $6 + 4 = \mathbf{10}$. (Echoes Worked Example 1.)
2. Add the cup back: $2 + 1 = 3$; that 3 is half the jug, so double: **6 cups**. (Echoes Worked Example 3 — reverse order.)
3. Doubling four times is undone by halving four times: $80 \rightarrow 40 \rightarrow 20 \rightarrow 10 \rightarrow \mathbf{5 \text{ grams}}$. (Like Worked Example 2, repeated undo — just doubling instead of halving.)
4. A longer chain: $20 + 4 = 24$, $24 \div 2 = 12$, $12 - 5 = 7$.
5. Nobody says “find the start,” but that’s the question. Undo last-first: 7 got on, so $19 - 7 = 12$; 4 got off, so $12 + 4 = \mathbf{16 \text{ people}}$.
6. Undo last-first: add the \$8 back, $40 + 8 = 48$; tripling is undone by dividing by 3, $48 \div 3 = \mathbf{\$16}$. Adding the 8 back *before* dividing is the whole game.

What to watch for. Two slips. First, **undoing in the wrong order** — on problems 2 and 6, doing the multiply/divide before the add/subtract gives a wrong answer; point back to Worked Example 3. Second, **guess-and-check instead of undoing** — it can stumble onto the answer on problem 1, but it bogs down on 3, 4 and 6, which is exactly where the technique earns its keep. And a bare number is not the finished answer here: ask them to show the undo chain, the way the worked examples do.

How to run it. Read the hook, the chain picture, and all three worked examples out loud together before they touch problem 1. The problems they do solo, in one sitting or two. This is a learning packet, not a test: helping on a stuck problem is fine, but help by pointing back at the matching worked example, not at the answer.